

CS 171: Discussion Section 5 (10/6)

1 MACs and Pseudorandom Functions

In the construction of a fixed-length MAC that we saw in lecture (and in construction 4.5 in the textbook), Mac is a pseudorandom function. However we will show that this feature is not necessary.

Construct a secure deterministic MAC for n -bit messages such that Mac is not a pseudorandom function. Note: you may use a pseudorandom function $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ in your construction.

2 Difference Between Regular and Strong Security for MACs

Construct a MAC $\text{MAC}' := (\text{Gen}', \text{Mac}', \text{Verify}')$ that is secure but not strongly secure. In your construction, you may start with a secure MAC, $\text{MAC} := (\text{Gen}, \text{Mac}, \text{Verify})$.